

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 9: Soil

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	9 — Soil
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Paper 5 (hardest tier). Expert traps: chaining percolation-rate arithmetic across unequal volumes AND unequal times where the slower-finishing column is actually the faster-draining one once both are normalised; correcting depth/cross-section confounds so a faster drain time never alone proves coarser texture; tracing washed-down clay and iron from the A horizon's loss to the B horizon's gain while the C horizon stays barely organic and the bedrock contributes none; never letting 'more water always helps' survive once aeration and oxygen-starved roots enter the picture; separating weathering (break-in-place) from erosion (carry-away) even when both act on the same slope; reading water-retention-versus-crop-suitability tables where the highest-retaining soil is the wrong pick; and refusing to treat percolation rate and water-holding capacity as the same property when they are inverse. Do not write on this paper — mark answers on your answer sheet.

1. On a single mountain, three identical granite outcrops sit at the foot (warm, wet), mid-slope (mild, moist) and summit (cold, dry). After ten thousand years the foot has the deepest soil and the summit almost none. A student says 'the foot rock must simply have been softer to begin with.' The soundest correction is that:

- (A) the summit rock was hardest from the start, which is the only reason it resisted forming soil
- (B) altitude by itself always produces the deepest soil at the highest point of any mountain

(C) the rock began identical, but the foot's warmth and abundant moisture sped both the chemical and physical breakdown, so climate, not original hardness, set the depth order

(D) the foot soil was carried up as ready-made soil by rising groundwater rather than weathered in place

2. A volcanic island is dated: the youngest lava flow (50 years) has a dusting of soil, a 600-year flow has a thin layer, and a 6000-year flow has deep dark soil. A tourist says 'so one wet monsoon could have built that deep soil in a season.' This is wrong because:

(A) deep soil here built up only as rock slowly weathered and successive generations of plants added humus across thousands of years, not within one season

(B) a single monsoon does build that depth, since the youngest 50-year flow already shows finished deep soil

(C) monsoon water carries pre-formed deep soil and drops it on lava within days

(D) the deepest soil always forms on the newest flow and then thins as the flow ages

3. A student lists processes and asks which is weathering AND which is erosion: (i) frost splitting a boulder where it lies, (ii) monsoon runoff sweeping topsoil down a bare slope, (iii) wind-blown sand scouring a cliff face, (iv) a river carrying loosened gravel to the sea. Grouping them correctly gives:

(A) all four are erosion, because each one involves moving rock or soil from where it formed

(B) (i) and (ii) are weathering while (iii) and (iv) are erosion, sorting them simply by whether wind is involved

(C) all four are weathering, because each one ultimately helps turn rock into soil

(D) (i) and (iii) are weathering (rock broken in place), while (ii) and (iv) are erosion (loosened material carried away)

4. Two neighbouring plots share rock and climate. Plot 1 kept under dense forest 300 years; plot 2 cleared and cropped bare for 120 years. Plot 1 has a thick dark surface band; plot 2's surface is now a pale firm layer faintly stained with clay. Reasoning backward, plot 2's pale surface is most likely:

(A) a thicker, paler version of the A horizon that simply lost its dark colour with age

(B) an exposed B horizon, revealed after erosion stripped the humus-rich A horizon that bare cropping failed to protect

(C) freshly risen bedrock that weathered to a stained clay layer within the 120 years

(D) a brand-new O horizon of half-rotted leaves formed by the crops grown there

5. A pit shows top to bottom: a dark crumbly band, a paler firm band stained with clay and iron, loose rock chips, then solid rock. A student says 'the clay and iron in the stained band were made there by the band's own chemistry.' The better account is that the clay and iron:

- (A) were leached out of the dark band above by downward-seeping rainwater and deposited in the stained band, which is the zone of accumulation
- (B) rose by capillary action from the solid rock at the base into the stained band
- (C) were blown in sideways from a neighbouring field and settled only in that one band
- (D) were carried upward from the loose rock chips by earthworms feeding at depth

6. Ranking the four profile layers by how much washed-down clay and iron they have GAINED from elsewhere, from most to least, the best order is:

- (A) A horizon, then B horizon, then C horizon, then bedrock
- (B) B horizon, then A horizon, then C horizon, then bedrock
- (C) bedrock, then C horizon, then B horizon, then A horizon
- (D) C horizon, then B horizon, then A horizon, then bedrock

7. A contractor levels a building site by trucking in 30 cm of pale, firm clay-stained subsoil and spreading it over an old garden's dark topsoil. The new owner plants vegetables directly into the imported layer and they grow poorly. The buried-profile reason is that the surface is now:

- (A) B-horizon subsoil low in humus and nutrients, sitting on top of the buried fertile A horizon the roots cannot reach
- (B) the most fertile A horizon, simply re-laid, so the poor growth must be due to weather alone
- (C) solid bedrock, which is why nothing can weather or grow there any more
- (D) an O horizon of fresh leaf litter that has not yet rotted into usable humus

8. A claim states: 'The C horizon is the richest in humus because it is closest to the parent rock that feeds the soil.' Judged against how a profile forms, this claim is:

- (A) wrong — the C horizon is partly weathered parent rock with very little humus, while humus is richest in the A horizon at the top
- (B) correct, because nutrients flow upward from rock so the deepest soil layer holds the most humus
- (C) correct, because the C horizon is actually the dark surface topsoil under another name
- (D) wrong only because the C horizon is made entirely of living organisms rather than rock

9. A soil scientist labels three jars from one pit: jar X = darkest, most crumbly, full of fine roots; jar Y = pale, firm, stained with washed-down clay; jar Z = loose chips of barely-changed parent rock. From top of the pit downward, the jars were filled in the order:

- (A) Z first, then Y, then X, because rock fragments always lie at the very top of a pit
- (B) X (A horizon) first, then Y (B horizon), then Z (C horizon)
- (C) Y first, then X, then Z, because the clay-stained layer is the surface topsoil
- (D) X first, then Z, then Y, because parent-rock chips sit directly under the topsoil

10. A desert profile shows almost no dark surface band; a rainforest profile shows a deep one, though both have the same bedrock. The desert's missing dark band is best explained because:

- (A) frequent heavy desert rain constantly washes the dark layer down into the bedrock
- (B) desert sand sits on no bedrock, so no profile can develop there at all
- (C) sparse plant life supplies very little litter, so almost no humus forms to darken and build up the surface layer
- (D) desert heat melts the humus, which then evaporates away each summer

11. Column 1 drains 480 mL in 8 minutes; column 2 drains 360 mL in 9 minutes. Comparing percolation rates, which column drains faster and what are the two rates?

- (A) column 1 is faster: 60 mL/min versus column 2's 40 mL/min
- (B) column 2 is faster, because it ran for the longer time of 9 minutes
- (C) column 1 is faster at 480 mL/min versus 360 mL/min, comparing the volumes directly
- (D) they are equal, since both drained a few hundred millilitres

12. A gardener records that bed A lets 540 mL through in 9 minutes while bed B lets 720 mL through in 18 minutes, then claims 'bed B drains faster because more water came out of it.' Computing the rates, the truth is that bed A's rate exceeds bed B's by the factor:

- (A) they match exactly, because the larger 720 mL volume of bed B cancels its longer time
- (B) 1.5 (bed A is 60 mL/min, bed B is 40 mL/min, so A is one-and-a-half times faster)
- (C) 2, obtained by dividing bed B's 18 minutes by bed A's 9 minutes
- (D) 0.75, since bed A's 540 mL is three-quarters of bed B's 720 mL

13. Three equal-section columns of the SAME loamy soil are poured with 600 mL each but at different depths. The 10 cm column drains in 5 min, the 20 cm in 10 min, the 30 cm in 15 min. A student concludes the deepest column is 'the clayey one.' The flaw is that:

- (A) all three are the same loam; the longer times simply reflect greater depth the water must cross, not a clayey texture
- (B) the deepest column must be clayey, because taking the longest time always proves finer particles
- (C) depth has no effect on drain time, so the differing times must mean three different textures
- (D) the shallowest column is clayey, because finishing fastest always proves the smallest pores

14. A fair-test failure: a student pours 400 mL on a sandy column but 200 mL on a clayey column, times both, and reports the clay's longer time as 'proof clay drains slower.' The conclusion is unreliable chiefly because:

- (A) percolation cannot be measured at all for clayey soil, so the test is meaningless
- (B) the two soils should have been mixed before any water was poured

(C) the poured volume was not kept equal, so the raw times cannot be compared unless rates (volume divided by time) are computed instead

(D) the timer must be started only after each column has finished draining

15. Soil P drains 600 mL in 12 minutes; soil Q drains 450 mL in 9 minutes; soil R drains 800 mL in 20 minutes. Listing them from highest percolation rate to lowest gives:

(A) R, then P, then Q, because R let the largest total volume through

(B) P, then Q, then R, in order of the volumes poured

(C) R, then Q, then P, ranking by the longest drain time first

(D) Q (50 mL/min), then P (50 mL/min) tie, then R (40 mL/min) — Q and P tie at 50, R lowest

16. An irrigation column percolates at a steady 24 mL/min. The first test pours 600 mL through it; the second pours 900 mL. The two drain times are:

(A) 25 minutes for 600 mL and 37.5 minutes for 900 mL

(B) 24 minutes for 600 mL and 36 minutes for 900 mL, reading the rate straight off as the time

(C) 14400 minutes for 600 mL, by multiplying the rate by the volume

(D) 0.04 minutes for 600 mL, by dividing the rate by the volume

17. A reservoir of 1000 mL is poured onto a steady column that drains at the rate shown by its first reading: 150 mL gone after 3 minutes. How much of the 1000 mL is left undrained after a further 7 minutes (10 minutes total)?

(A) 850 mL remain, since only the first 150 mL ever drains and the rest stays put

(B) 500 mL remain (rate 50 mL/min, so 500 mL drained in 10 minutes)

(C) 0 mL remain, because the whole 1000 mL must drain within the first 10 minutes

(D) 950 mL remain, taking the 50 mL/min rate itself as the total volume drained

18. A lab pours equal 720 mL onto two pots that are filled to exactly the same depth and width. Pot 1 finishes in 12 min, pot 2 in 16 min. The lab reports the gap between their drainage rates as '4 units' by subtracting the times. Correcting this, the actual rate gap and the sandier pot are:

(A) a gap of 15 mL/min (pot 1 at 60, pot 2 at 45), with pot 1 the sandier soil

(B) a gap of 4 mL/min, since subtracting 12 from 16 directly gives the rate difference

(C) no gap at all, because identical depth and equal 720 mL force identical rates

(D) a gap of 15 mL/min, but with pot 2 the sandier soil because it drained for longer

19. A managed field with added humus drains 540 mL in 6 min; an adjacent compacted, over-cultivated field drains the same 540 mL in 18 min. The looser humus-rich structure most directly explains why the managed field has the:

(A) lower percolation rate, because humus clogs the pores and slows the water down

(B) same rate as the compacted field, since both drained the identical 540 mL

- (C) higher rate of 90 mL/min only because its particles are larger grains of sand
- (D) higher percolation rate, 90 mL/min versus 30 mL/min, because its more open pore spaces let water pass faster

20. A student suspects soil X is loam and soil Y is clay. X drains 480 mL in 8 min; Y drains 480 mL in 32 min. The sound conclusion is:

- (A) Y drains faster, which would make Y the loam
- (B) the rates are equal because both samples passed the same 480 mL
- (C) X's rate (60 mL/min) is four times Y's (15 mL/min), consistent with X being the freely-draining loam and Y the slow clay
- (D) X is the clay, because its higher rate means it holds water longer

21. A farmer measures plot A at 6 mL/min and plot B at 54 mL/min. He must site paddy (likes standing water) and groundnut (likes loose, well-drained ground). The best placement is:

- (A) paddy in plot A, whose slow 6 mL/min drainage holds standing water, and groundnut in plot B, whose fast 54 mL/min drainage gives loose, airy soil
- (B) paddy in plot B, because rice grows best where water drains away fastest
- (C) groundnut in plot A, because groundnut needs the same standing water that rice does
- (D) either crop in either plot, since percolation rate does not affect crop choice

22. Four soils are tested with DIFFERENT volumes: W passes 200 mL in 5 min, X passes 360 mL in 4 min, Y passes 150 mL in 10 min, Z passes 480 mL in 8 min. The soil most likely to be clay (the slowest-draining) is:

- (A) soil X, because it took the shortest time of only 4 minutes
- (B) soil Z, because it passed the largest total volume of 480 mL
- (C) soil W, because it passed the smallest total volume of 200 mL
- (D) soil Y, whose rate of 15 mL/min is the lowest of the four

23. A student weighs 100 g of dark garden soil, dries it gently to remove water (now 90 g), then heats it strongly until it stops changing and turns pale (now 82 g). The 90-to-82 g drop in the SECOND step is best explained by:

- (A) more water leaving, even though the gentle drying had already removed all the water
- (B) the burning away of humus (organic matter), which is also why the soil turned pale
- (C) the sand and silt grains burning away in the strong heat
- (D) oxygen being added to the soil, which somehow lowered its weight

24. Three 100 g moist field samples are gently heated to dryness and re-weighed: sample A now 92 g, sample B now 78 g, sample C now 85 g. Assuming the loss is mainly water, which sample most likely came from clayey soil, and why?

- (A) sample A, which lost the least mass, because clay holds the least water of the three
- (B) sample C, the middle loss, because clay always holds a moderate amount of water
- (C) sample B, which lost the most mass (22 g of water), since clay's high water-holding capacity means it had retained the most moisture
- (D) none of them, because heating cannot reveal which soil held the most water

25. A dry clod dropped in water streams bubbles; a second clod, pressed hard into a tight pellet first, gives far fewer bubbles. A classmate says 'pressing destroyed the soil's water.' The correct reading is that pressing:

- (A) removed the soil's water, leaving nothing left to form any bubbles
- (B) made the clod absorb air out of the surrounding water in the beaker
- (C) closed the pore spaces between particles, so less air was trapped to escape as bubbles
- (D) proved that only soil containing no air at all can ever release bubbles

26. A student runs two soil experiments side by side: experiment 1 drops a dry clod into water and bubbles rise; experiment 2 heats a moist clod in a tube and mist condenses on the cool glass. Matching each result to what it proves the soil contains:

- (A) both experiments prove only that the soil holds air, since bubbles and mist are both just gas
- (B) experiment 1 proves water and experiment 2 proves air, the two conclusions reversed
- (C) neither proves anything, because soil contains neither air nor water until rain adds them
- (D) experiment 1 proves the soil holds air (escaping as bubbles) and experiment 2 proves it holds water (vapour recondensing as mist)

27. A soil after rain is, by volume, about 45% mineral grains, 5% humus, 25% water and 25% air. The same sample is then baked dry. Which pair of changes best describes what happens to the water and the air?

- (A) the mineral grains evaporate while the water stays exactly as it was
- (B) the humus boils off entirely and the air disappears completely as the soil dries
- (C) the water largely evaporates, and the space it vacates is taken up by more air
- (D) the air turns into water as the soil dries, swelling the water fraction

28. A lab measures three soils for the SAME 300 mL test: soil with the largest grains drains in 4 min, the balanced humus-rich mix in 8 min, the finest tightly-packed soil in 20 min. If a fourth soil drains the 300 mL in 6 min, its likely texture is:

- (A) clayey, because 6 minutes is the longest time and so the slowest of all
- (B) between sandy and the loamy mix — coarser than the loam, since 6 min is faster than the loam's 8 min but slower than the pure sand's 4 min

- (C) the same as the finest tightly-packed soil, which also took 6 minutes
- (D) impossible to judge, because percolation time tells nothing about texture

29. A field with many earthworm burrows is compared with an identical field where worms were killed by chemicals. Of the changes below, which is the one most directly expected in the worm-free field, and why?

- (A) a lower percolation rate, because without worm burrows fewer open channels remain for water to seep down
- (B) a higher percolation rate, because worms had been clogging the channels with their castings
- (C) no change in percolation, because worms affect only the soil's colour, not its water movement
- (D) a faster percolation rate plus better aeration, because removing worms loosens the soil further

30. A seed company advertises a soil mix 'engineered to both drain instantly AND stay the wettest of any soil.' An expert calls the advert physically impossible because:

- (A) draining instantly and staying wettest are the same property, so the mix simply does both at once
- (B) any soil with enough humus added will automatically drain fastest and hold the most water together
- (C) clay alone achieves both, since its slow drainage is what makes it drain instantly
- (D) instant drainage needs wide pores between coarse grains, but staying wettest needs tiny pores that hold water — one feature undoes the other

31. After one downpour, a gardener notes the order in which three beds become walkable (firm, not soggy): bed 1 first, then bed 2, then bed 3 last. If the beds are sand, loam and clay in some order, the matching most consistent with their properties is:

- (A) bed 1 is clay, bed 2 is loam, and bed 3 is sand, since clay sheds water fastest
- (B) bed 3 is sand, because sand holds the most water and so stays soggy the longest
- (C) bed 1 is sand, bed 2 is loam, and bed 3 (last to firm up) is clay
- (D) all three firmed up together, so the beds must all be the same texture

32. Four pots are set up: (1) a flooded clay pot left standing in a tray of water, (2) a flooded sandy pot whose tray has been emptied so it drains freely, (3) a moist freshly-dug loam pot, (4) a dry sandy pot. Ranking root aeration from WORST to best, the worst pot is:

- (A) pot 2, the freely draining flooded sand, because any pot that was ever flooded can never re-admit air
- (B) pot 4, the dry sandy pot, because dry soil contains no air for roots at all
- (C) pot 1, the flooded standing clay, whose fine pores stay water-filled so almost no air reaches the roots
- (D) pot 3, the moist loam, because moist soil always holds less air than bone-dry soil

33. A gardener has two problem beds: bed P is pure sand that dries out within hours; bed Q is heavy clay that stays water-logged for days. He has a pile of well-rotted compost. Used wisely, the compost will:

- (A) help both beds — adding it to the sandy bed raises its water retention, and adding it to the clay bed opens its structure to drain a little better
- (B) help only the clay bed, since compost can never improve a sandy soil's water retention
- (C) harm both beds by sealing their pores so neither drains nor holds water any more
- (D) turn the sandy bed instantly into solid clay and the clay bed into loose sand

34. Four students each offer one piece of evidence that soil is 'teeming with life.' Whose evidence is the only one that actually demonstrates living organisms rather than a non-living property?

- (A) the student who shows the soil bubbles when dropped in water, proving it is alive
- (B) the student who counts millions of micro-organisms plus visible mites and worms under a lens in one gram of soil
- (C) the student who shows the soil turns a test tube's walls misty when heated, proving it breathes
- (D) the student who shows the soil loses mass when strongly heated, proving organisms burned off

35. A salt-crusted sandy patch supports very few organisms; a humus-rich loamy patch teems with them. The loam supports far more life chiefly because it offers:

- (A) the largest particles, which most soil organisms strongly prefer
- (B) the fastest drainage, so the organisms are never disturbed by water
- (C) the coldest temperature, since colder soil always supports the most life
- (D) moisture, humus as food, and air pockets — the combination soil organisms need

36. A potter explains why a thrown clay bowl can briefly hold water while a sandcastle cannot. The same property of clay that lets the bowl hold water is the one that, in a field, makes clay soil:

- (A) drain very fast, because the same tight packing forces water straight through
- (B) drain very slowly, because clay's tiny tightly-packed pores resist the downward flow of water
- (C) hold no water at all, because tightly packed clay leaves no space for moisture
- (D) have the same percolation rate as sand, since both are just made of mineral grains

37. Two adjacent clay plots grow gram (chickpea). Plot A is watered moderately and drains between waterings; plot B is kept under standing water 'to be generous.' Plot B's gram wilts and its roots rot while plot A thrives. The key difference is that plot B's standing water:

- (A) kept the clay's fine pores full of water, excluding air, so the roots could not get oxygen and rotted
- (B) drained away so fast through the clay that plot B's roots dried out and died

(C) washed all the nutrients up to the surface where the gram roots could not reach them

(D) proved gram is a water crop, so plot A must actually have been under-performing

38. An advisor matches crops to four soils: paddy, cotton, gram and groundnut must each go to one of a water-logging clay, a moisture-holding clayey-loam, a light sandy-loam, and a fast-draining sand. The fully correct matching is:

(A) paddy to the fast-draining sand, cotton to the water-logging clay, gram to the sandy-loam, groundnut to the clayey-loam

(B) groundnut to the water-logging clay, paddy to the sandy-loam, cotton to the clay-loam, gram to the sand

(C) all four crops to the water-logging clay, since heavy soils suit every crop

(D) paddy to the water-logging clay, gram to the moisture-holding clayey-loam, cotton to the light sandy-loam, groundnut to the fast-draining sand

39. A school garden tests three soils for a tomato bed that must stay moist yet drain after rain: soil J drains in 3 min and dries by noon; soil K drains in 9 min and stays evenly moist; soil L drains in 40 min and stays soggy two days. The best pick and reason is:

(A) soil J, because the fastest-draining soil is always the healthiest for any crop

(B) soil L, because the soil that stays wettest longest gives tomatoes the most water they need

(C) soil L, because its very slow drainage guarantees the roots are always well supplied with air

(D) soil K, whose middling drainage keeps the bed evenly moist while still shedding the surplus after rain

40. An agronomist studies two failing fields. Field 1: thin dark topsoil over thick coarse sand; crops keep wilting from drought even after watering. Field 2: thick dark topsoil over deep heavy clay; crops keep rotting after rain. The single different cause in each case is:

(A) both fields fail for the same reason — too much humus making the soil water-logged

(B) field 1 water-logs from its sand and field 2 dries out from its clay, the sand and clay each acting backwards

(C) field 1 loses water to fast percolation below the roots, while field 2 water-logs as the clay holds water and excludes air

(D) both fail because their topsoils have an unusually high water-holding capacity that dries the crops out

41. Two students debate. One says 'sand drains fastest, so sand is the best soil for every crop.' The other says 'clay holds water, so clay is the best soil for every crop.' The correct judgement of both claims is:

(A) the first student is right — the fastest-draining soil is genuinely best for all crops

(B) the second student is right — the most water-retentive soil is genuinely best for all crops

(C) both are right at once, because the best soil for every crop is whichever drains fastest and holds the most water together

(D) both are wrong — neither soil is best for every crop; sand suits drought-tolerant crops like groundnut, while clay suits water-loving crops like paddy

42. A farmer owns two fields: field G is light, fast-draining sandy-loam; field H is heavy, moisture-holding clayey-loam. He must place paddy, cotton and gram. With only two fields, the best plan is:

(A) paddy in field G, because rice grows best where water drains away fastest

(B) all three crops in field G, since the sandy-loam suits every crop equally well

(C) cotton in the sandy-loam field G, and paddy or gram matched to field H's moisture-holding clayey-loam by their water needs

(D) cotton in field H, because cotton thrives only on permanently water-logged clay

43. Four things happen to a field over a decade: (i) a factory drains toxic effluent onto it, (ii) plastic bags accumulate and choke the soil, (iii) monsoon runoff strips its topsoil downhill, (iv) fallen leaves rot into dark humus. Which set counts as soil pollution, as opposed to erosion or natural enrichment?

(A) (iii) only, because washing topsoil downhill is the clearest form of soil pollution

(B) (i) and (ii), because adding toxic effluent and plastic poisons or chokes the soil, while (iii) is erosion and (iv) is natural enrichment

(C) (iv) only, because rotting leaves chemically alter the soil and so pollute it

(D) all four, since any change to a field counts as pollution of its soil

44. Two hill farmers fight erosion differently. Farmer 1 ploughs furrows straight up and down the slope; farmer 2 cuts level terraces and ploughs along the contour. After the monsoon farmer 1 has deep gullies and farmer 2 has little soil loss. The reason farmer 2's method works is that it:

(A) breaks and slows the downhill flow of water so it soaks in rather than racing down and carrying soil away

(B) channels the water downhill even faster, flushing the loose soil away before it can erode

(C) makes the rainwater evaporate off the terraces before it can touch the soil at all

(D) removes all the soil into the terraces ahead of the rains so none is left to erode

45. An officer must recommend ONE erosion measure for each of two sites: a flat dry windswept plain losing dust, and a steep rainy hillside losing soil in gullies. The correct pairing of measure to site is:

(A) terracing for the flat plain, and shelter belts for the steep hillside, swapping the two

(B) deep ploughing for both sites, since turning the soil over always stops erosion

(C) shelter belts of trees for the windswept plain, and terracing with contour ploughing for the steep rainy hillside

(D) leaving both sites bare, so neither wind nor rain has any standing crop to damage

46. A student models soil with two jars of equal-sized beads: jar 1 holds large marbles, jar 2 holds fine sand grains. Water poured in jar 1 runs straight through; in jar 2 it trickles slowly. Mapping this to real soils, jar 1 behaves like sand and jar 2 like clay because:

- (A) the marbles dissolve in the water and open extra channels that the sand grains cannot
- (B) the fine grains chemically push the water back up out of the jar against gravity
- (C) the marbles have no gaps between them at all, so water simply slides over the top
- (D) larger particles leave wider gaps that pass water quickly, while finer particles leave narrow gaps that resist flow

47. On one hillside, frost first cracks a granite boulder into rubble over decades; then a single flood sweeps that rubble down to the valley in an afternoon. A student labels the whole event 'erosion.' The precise correction is that:

- (A) the whole event was weathering, because rock was being broken down throughout
- (B) the frost cracking the boulder in place was weathering; only the flood carrying the rubble away was erosion
- (C) the frost cracking was erosion and the flood transport was weathering, the two reversed
- (D) neither stage counts, since weathering and erosion both require wind rather than frost or water

48. A road-cut exposes the same granite at two spots 20 km apart: one in a hot humid valley with lush forest, the other on a cold dry windswept ridge with bare rock. A geologist predicts which will have the thicker soil and why. The sound prediction is:

- (A) the cold dry ridge, because cold dry air dissolves rock faster than warm humid air does
- (B) the humid forested valley, because warmth and moisture speed rock weathering while abundant litter decays into deep humus
- (C) the cold dry ridge, because its higher, windier position always builds the deepest soil
- (D) they will be identical, because the granite is the same so the soil must be the same depth

49. Four chunks of the same sandstone are left fifteen years: chunk 1 in a riverbed where water repeatedly freezes in its cracks each winter; chunk 2 in a sealed dry display case; chunk 3 buried under bone-dry desert sand at steady temperature; chunk 4 indoors at constant room temperature with dry air. Which chunk crumbles MOST, and why?

- (A) chunk 1 in the freezing riverbed, because water expanding as it freezes in the cracks repeatedly pries the rock apart
- (B) chunk 2 in the sealed dry case, because enclosing a rock with no air movement speeds its break-up

(C) chunk 3 buried in dry desert sand, because being buried always crushes rock to fragments fastest

(D) chunk 4 indoors, because steady room temperature with dry air weathers rock the fastest of all

50. A geologist distinguishes soil from bedrock by their relation to life. The statement that captures the difference correctly is:

(A) bedrock contains the humus and organisms, while soil is the lifeless rock layer above it

(B) both soil and bedrock are equally full of humus and organisms throughout

(C) soil is pure organic matter only, with no mineral content from weathered rock at all

(D) soil is weathered mineral material mixed with humus and living organisms, whereas bedrock is solid unweathered rock essentially without those living components

51. After years of heavy rain, a profile's A horizon has grown noticeably paler and thinner while the B horizon below has thickened with extra clay and iron. A student asks where the B horizon's new material came from. The correct answer is:

(A) it rose up from the solid bedrock by capillary action into the B horizon

(B) it was leached downward out of the A horizon by seeping rainwater, which is why the A horizon paled and thinned as the B thickened

(C) it was blown in from a neighbouring field and settled only in the B horizon

(D) the B horizon manufactured the clay and iron from the air trapped in its pores

52. Soil S drains 360 mL in 4 minutes; soil T drains 360 mL in 12 minutes; soil U drains 360 mL in 6 minutes. A student says 'U is sandiest because 6 is the smallest middle number.' The correct ranking from sandiest to clayiest is:

(A) U sandiest, then S, then T, because 6 minutes feels like the balanced middle

(B) T sandiest, then U, then S, because the longest time means the loosest soil

(C) S (90 mL/min) sandiest, then U (60 mL/min), then T (30 mL/min) clayiest

(D) all three are the same texture, since each drained the identical 360 mL

53. A column drains at a steady rate, passing 240 mL in 6 min. The experimenter wants to know two things: the rate, and the time to pass 600 mL at that same rate. The correct pair is:

(A) rate is 40 mL/min, but 600 mL takes 24000 minutes

(B) rate is 40 mL/min, and 600 mL takes 15 minutes

(C) rate is 1440 mL/min, and 600 mL takes about 0.4 minutes

(D) rate is 6 mL/min, and 600 mL takes 100 minutes

54. Two columns of the SAME soil are compared. Column 1 is 20 cm deep and drains 500 mL in 10 min; column 2 is 10 cm deep and drains 500 mL in 5 min. A student concludes 'column 2 is sandier.' The flaw is that:

- (A) the deeper column must be sandier, since taking longer proves coarser grains
- (B) soil depth has no effect at all on how long percolation takes
- (C) a faster drain time always proves a larger particle size regardless of the setup
- (D) both columns hold the same soil, and column 2's shorter time mainly reflects the smaller depth the water had to cross, not a coarser texture

55. Three nearby plots share climate and rock. Plot 1: undisturbed forest 250 years. Plot 2: cropped but mulched and rotated 60 years. Plot 3: cropped bare with stubble burned 60 years. Their A-horizon thicknesses, from thickest to thinnest, are most likely:

- (A) plot 3 thickest, then plot 2, then plot 1, because cropping and burning build topsoil fastest
- (B) all three equal, because the same climate and rock must give the same A horizon
- (C) plot 2 thickest, then plot 1, then plot 3, because mulched cropland always beats forest
- (D) plot 1 thickest, then plot 2, then plot 3 thinnest, since organic input and protection decline across the three

56. Two identical bare plots are compared one monsoon: plot 1 is left bare, plot 2 is covered with a layer of straw mulch. Plot 1 loses far more topsoil and dries out faster between rains. The mulch helped plot 2 chiefly by:

- (A) making plot 2's soil so heavy that rain and wind could not lift any of it
- (B) sealing plot 2's surface so completely that no rain could percolate in at all
- (C) hardening plot 2's topsoil into bedrock that erosion could no longer touch
- (D) breaking the force of falling rain and wind on the surface while holding moisture, so the topsoil stayed in place and damp

57. A footpath is worn diagonally across a loamy lawn. After rain, puddles form along the path while the untrodden lawn beside it drains freely, though both are the same soil. Measuring percolation, the path strip would show a:

- (A) higher percolation rate than the lawn, because the packed path lets water through faster
- (B) rate identical to the lawn, because trampling cannot change the same soil's drainage
- (C) lower percolation rate than the lawn, because trampling compacted its pores closed so water seeps in more slowly
- (D) higher rate, because compaction enlarges the pores and so speeds the water through

58. A timeline lists events on an exposed rock outcrop: (i) lichens and roots slowly prise grains loose over centuries, (ii) winter ice wedges open its cracks year after year, (iii) a one-second lightning strike chars a patch, (iv) wind-blown sand polishes its face over decades. Which events are the genuine long-term weathering agents that build soil parent material?

- (A) (iii) alone, because a single lightning strike does more weathering than centuries of the others
- (B) (i), (ii) and (iv) — biological prising, freeze-thaw and wind abrasion all break rock down slowly over long periods
- (C) (ii) and (iii), because only sudden, violent events such as ice cracking and lightning weather rock
- (D) none of them, because rock turns to soil only by suddenly melting, not by slow weathering

59. A profile shows, top to bottom: spongy half-rotted leaves, a dark crumbly band, a pale clay-stained firm band, loose rock chips, then solid rock. Which layer is correctly matched to its identity?

- (A) the pale clay-stained firm band is the B horizon, where leached clay and iron accumulate
- (B) the dark crumbly band is the C horizon of barely-weathered parent rock
- (C) the loose rock chips are the A horizon richest in humus
- (D) the spongy half-rotted leaves are the solid bedrock at the base

60. Two soils are tested for a fair comparison with equal 720 mL volumes. Soil X finishes draining in 9 minutes; soil Y finishes in 15 minutes. A student says 'soil Y is faster because it kept draining for longer.' The correct rates and conclusion are:

- (A) soil Y is faster at 48 mL/min, because draining over more minutes always means a higher rate
- (B) the rates are equal because both soils passed the same 720 mL of water
- (C) soil X is faster at 80 mL/min versus soil Y's 48 mL/min, so the longer drain time of Y means slower, not faster
- (D) soil X drains at 6480 mL/min, found by multiplying the volume by the time

Solutions — Science (Class 7), Chapter 9: Soil — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	A	21	A	31	C	41	D	51	B
2	A	12	B	22	D	32	C	42	C	52	C
3	D	13	A	23	B	33	A	43	B	53	B
4	B	14	C	24	C	34	B	44	A	54	D
5	A	15	D	25	C	35	D	45	C	55	D
6	B	16	A	26	D	36	B	46	D	56	D
7	A	17	B	27	C	37	A	47	B	57	C
8	A	18	A	28	B	38	D	48	B	58	B
9	B	19	D	29	A	39	D	49	A	59	A
10	C	20	C	30	D	40	C	50	D	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) All three outcrops are the same granite, so original hardness cannot explain the gradient. The decisive variable is climate: warmth and plentiful water at the foot accelerate both chemical dissolution and physical (freeze-thaw, root) breakdown of rock, while the cold dry summit weathers very slowly. Hence depth tracks environment, not a pre-existing softness. The claim that the summit rock was hardest is contradicted by 'identical' rock. Altitude does not deepen soil at the top; here the high site has the least. Groundwater does not lift finished soil upward; soil forms in place as rock weathers and organisms add humus.

2. (A) The dated series shows depth increasing with age over millennia — soil deepens as rock weathers and humus accumulates slowly. One season cannot replicate thousands of years of weathering and litter decay. The 50-year flow has only a dusting, not finished soil, so the 'one monsoon' claim is refuted by its own data. Floods do not deposit ready-made deep soil. The youngest flow has the least soil, so depth does not start high and thin with age — it is the reverse.

3. (D) Weathering breaks rock in place into smaller pieces; erosion is the carrying-away of loosened material. Frost-splitting (i) and wind-scouring (iii) break rock where it sits, so both are weathering even though (iii) uses wind. Runoff sweeping topsoil (ii) and a river transporting gravel (iv) both remove and transport material, so both are erosion. 'All erosion' wrongly reclassifies the in-place break-up. Sorting by 'wind involved' is a false rule — wind can do either job. 'All weathering' ignores that (ii) and (iv) are transport, not break-up.

4. (B) Forest litter kept plot 1's A horizon thick and dark while erosion was held in check. Bare cropping on plot 2 let the unprotected A horizon erode away; the layer now at the surface is firm and stained with clay washed down from above — the signature of the B (accumulation) horizon, now exposed. A 'paler A horizon' contradicts the clay-staining and loss of humus. Bedrock does not rise and weather to clay in 120 years. A fresh O horizon would be spongy half-rotted leaves, not a firm clay-stained layer.

5. (A) The dark crumbly band is the A horizon (zone of loss); the stained firm band is the B horizon (zone of gain). Rainwater seeps down through the A horizon, washing fine clay and dissolved iron out of it and depositing them in the B horizon below — so the staining is imported from above, not self-made. Capillary rise from bedrock does not supply the bulk clay-iron of the B horizon. Sideways wind delivery to a single buried band is not how leaching works. Earthworms move material but the dominant clay-iron transfer here is downward leaching, not upward worm carriage.

6. (B) Gain of leached clay and iron is greatest in the B horizon (the zone of accumulation). The A horizon is a zone of loss, but it still passes and holds a little before it migrates, so it ranks above the deep layers. The C horizon is barely-weathered parent rock with little imported fine material, and solid bedrock receives essentially none. Ordering A first reverses loss and gain. Putting bedrock or C first ignores that the deep unweathered layers are not where leached material collects.

7. (A) The imported pale, clay-stained, firm layer is B-horizon subsoil — low in humus and nutrients — and it now caps the original humus-rich A horizon, which is buried out of root reach, so vegetables struggle. It is not a re-laid A horizon; the A horizon is dark and humus-rich, not pale subsoil. It is loose subsoil, not solid bedrock. And clay-stained subsoil is not an O horizon of leaf litter, which would be spongy and organic.

8. (A) Humus forms from decaying plant and animal litter at the surface, so the A horizon is humus-richest; the C horizon is broken, partly-weathered parent rock with little organic matter. Rock does not pump humus upward into deep layers. The C horizon is not the surface topsoil — that is the A horizon. And C is mostly mineral rock fragments, not entirely living organisms, so that 'wrong' reasoning is itself incorrect.

9. (B) Top to bottom a profile runs A (dark, crumbly, root-filled topsoil = jar X), B (pale, firm, clay-stained accumulation = jar Y), then C (loose parent-rock chips = jar Z). So the

fill order is X, Y, Z. Rock fragments lie deep, not at the top, so Z is not first. The clay-stained layer is the buried B horizon, not the surface, so Y is not first. The C horizon (Z) lies below the B horizon, not directly under the topsoil, so X-Z-Y is wrong.

10. (C) A dark surface band is humus-rich topsoil, and humus needs a steady supply of decaying plant and animal litter. Deserts have sparse vegetation, so little litter and little humus form — hence the thin or absent dark band. Deserts are dry, not heavily rained on, so 'rain washing it down' is false. The profiles share bedrock, so 'no bedrock' is contradicted. Humus does not melt and evaporate; it simply fails to accumulate without litter.

11. (A) Rate = volume / time. Column 1: $480 / 8 = 60$ mL/min. Column 2: $360 / 9 = 40$ mL/min. So column 1 drains faster. Longer time does not mean faster — it is the rate that matters, not the clock. Comparing raw volumes (480 vs 360) ignores the differing times, so '480 mL/min' is a misuse of the figures. The two rates differ (60 vs 40), so they are not equal.

12. (B) Rate = volume / time. Bed A: $540 / 9 = 60$ mL/min. Bed B: $720 / 18 = 40$ mL/min. $60 / 40 = 1.5$, so bed A drains 1.5 times faster — the gardener's 'more water means faster' is wrong, since bed B simply ran longer. The larger volume does not cancel the longer time to make them match. Dividing the times ($18/9 = 2$) ignores the unequal volumes. Comparing the raw volumes (540 vs 720) is meaningless once rates are found.

13. (A) The columns are stated to be the same loam, so texture is constant. Drain time rises in step with depth (5, 10, 15 min for 10, 20, 30 cm) — exactly what greater path length predicts at one rate. So the longest time reflects depth, not clay. 'Longer time proves finer particles' ignores the depth confound. Depth clearly does affect drain time here, so claiming it has no effect is false. Finishing fastest reflects the least depth, not the smallest pores.

14. (C) Comparing raw drain times is only valid when the same volume is used; here 400 mL versus 200 mL breaks the fair test. The proper fix is to compute each rate (volume / time) and compare those, since rate normalises for volume. Clay's percolation is perfectly measurable, just slow. Mixing the soils would destroy the comparison entirely. The timer must run while water drains, not start after — starting late would measure nothing.

15. (D) Compute rates: P = $600 / 12 = 50$ mL/min; Q = $450 / 9 = 50$ mL/min; R = $800 / 20 = 40$ mL/min. So P and Q tie at 50 mL/min and R is slowest at 40. Largest total volume (R) does not mean fastest — R is actually the slowest once normalised. Ranking by poured volume or by longest time both ignore that rate is volume per minute, the only fair measure.

16. (A) Time = volume / rate. $600 / 24 = 25$ min; $900 / 24 = 37.5$ min. Reading 24 as the time ignores that 24 is a rate (mL per minute), not minutes. Multiplying rate by volume

(24 x 600) gives a meaningless 14400. Dividing rate by volume inverts the formula. Only volume divided by rate yields the correct times.

17. (B) Rate = $150 / 3 = 50$ mL/min. In 10 minutes total, drained = $50 \times 10 = 500$ mL, so $1000 - 500 = 500$ mL remain. The column does not stop after 150 mL — the rate is steady. There is no rule forcing all 1000 mL through in 10 minutes; at 50 mL/min that would need 20 minutes. And 50 is the per-minute rate, not the volume drained, so 950 remaining is wrong.

18. (A) Rates: pot 1 = $720 / 12 = 60$ mL/min; pot 2 = $720 / 16 = 45$ mL/min; gap = $60 - 45 = 15$ mL/min. Subtracting times ($16 - 12 = 4$) gives a time difference in minutes, not a rate gap in mL/min — that is the lab's error. Equal depth and volume with unequal times still give unequal rates, so 'no gap' is wrong. The faster pot (1) is the sandier; pot 2's longer drain means it is the slower, finer soil, not the sandier one.

19. (D) Managed: $540 / 6 = 90$ mL/min. Compacted: $540 / 18 = 30$ mL/min. The looser, humus-improved structure opens pore spaces, raising percolation, so $90 > 30$. Humus and good structure open pores rather than clog them, so the rate rises, not falls. Equal volume with very different times means unequal rates. The faster rate comes from open pore structure, not from the soil suddenly being coarse sand.

20. (C) X: $480 / 8 = 60$ mL/min. Y: $480 / 32 = 15$ mL/min. $60 / 15 = 4$, so X drains four times faster — consistent with X = loam, Y = clay. Y is the slower drainer, so Y is not the loam. Equal volumes over very different times give unequal rates. A higher percolation rate means water passes quickly and is NOT retained, so X (fast) is the loam, not the clay.

21. (A) Paddy needs water to stand, so it belongs in the slow-draining plot A (6 mL/min). Groundnut needs loose, well-drained, airy soil, so it belongs in the fast-draining plot B (54 mL/min). Putting paddy in the fast plot would let its needed water drain away. Groundnut in the slow plot would water-log its roots. Percolation rate is central to crop choice, so 'either crop anywhere' is wrong.

22. (D) With unequal volumes you must compute rates. $W = 200/5 = 40$; $X = 360/4 = 90$; $Y = 150/10 = 15$; $Z = 480/8 = 60$ mL/min. The lowest rate, 15 mL/min (Y), means slowest drainage = clay. Shortest time (X, 4 min) is the fastest, not the slowest. Largest volume (Z) does not mean slowest — its rate is high. Smallest volume (W) does not by itself mean clay; its rate of 40 is middling. Only the computed rate identifies the clay.

23. (B) Gentle drying removed the water (100 to 90 g). The further loss on strong heating (90 to 82 g), with the soil turning pale, is the dark humus burning off — paling confirms organic matter is gone. The water was already removed in the drying step, so it cannot account for the second drop. Mineral sand and silt do not burn away. Adding oxygen would not lower weight, and that is not what strong heating does to humus — it burns it off.

24. (C) Loss = 100 minus final mass: A lost 8 g, C lost 15 g, B lost 22 g. The largest water loss (B, 22 g) points to the soil that had retained the most moisture — clay, with its high water-holding capacity. Sample A lost the least, so it held the least water (more like sand), not clay. The middle loss (C) is loam-like, not the clay extreme. Heating to dryness and weighing the loss is precisely how retained-water content is compared, so 'cannot reveal' is wrong.

25. (C) Bubbles come from air trapped in the soil's pore spaces escaping when water displaces it. Pressing compacts the clod, closing pores, so less air is trapped and fewer bubbles form. The bubbles are air, not water, so 'destroyed the water' is irrelevant. The clod releases air to the water, it does not absorb air from it. And it is the presence of trapped air — not its absence — that produces bubbles.

26. (D) Bubbles rising from a clod dropped in water are trapped air escaping (experiment 1 = air); mist condensing on cool glass after heating is the soil's own water turning to vapour and back to liquid (experiment 2 = water). Mist is condensed water, not gas, so 'both prove air' is wrong. The conclusions are not reversed — bubbles mean air, mist means water. And soil naturally contains both air and water, so 'neither proves anything' is false.

27. (C) Drying evaporates the water; the pores it leaves are then occupied by air, so the air fraction rises as the water fraction falls. Mineral grains are solid and do not evaporate. Humus does not boil off merely on sun-drying, and air does not disappear — it expands to fill the vacated pores. Air does not turn into water; that is the reverse of what drying does.

28. (B) Sand (largest grains) drained fastest at 4 min, loam at 8 min, clay (finest) slowest at 20 min. A 6-min time sits between sand and loam, so the fourth soil is coarser than the loam but a touch finer than pure sand — a sandy-loam. 6 min is NOT the longest time (20 min is), so it is not the clay. The finest soil took 20 min, not 6, so they are not the same. Drain time at equal volume directly indicates texture, so it is judgeable.

29. (A) Worm burrows are open channels that admit air and let water percolate; remove the worms and those channels are lost, so percolation falls and aeration worsens in the worm-free field. Worms do not clog channels — they create them. Worms affect structure and water movement, not merely colour. And killing worms removes, not adds, loosening, so percolation and aeration drop rather than improve.

30. (D) Fast (instant) drainage requires wide pores between coarse particles; maximal wetness requires fine, tightly-packed pores that retain water. These structures are opposites, so a soil cannot maximise both — percolation rate and water-holding capacity are inversely related. They are not the same property. Humus improves balance but cannot make a soil simultaneously the fastest-draining AND the most retentive. Clay's slow drainage is the opposite of 'drains instantly,' so the clay claim is self-contradictory.

31. (C) Sand drains fastest and holds least, so it firms up first (bed 1); clay drains slowest and holds most, so it stays soggy longest (bed 3); loam is intermediate (bed 2). Clay does not shed water fastest — it is the slowest. Sand does not stay soggiest — it holds the least water. Different textures dry at different rates, so they could not all firm up together.

32. (C) Aeration is worst where pores stay full of water. Pot 1's clay, standing in water, keeps its fine pores saturated, starving roots of air — the worst. Pot 2's sand drains freely, so air re-enters its wide pores once the tray is emptied; being flooded once does not permanently exclude air. Pot 4's dry sand is full of air, not devoid of it. Pot 3's moist loam still holds plenty of air in its open pores. Only standing water in fine clay pores chokes off air.

33. (A) Well-rotted compost (humus) improves structure in both directions: in pure sand it boosts water-holding capacity (and nutrients), and in heavy clay it opens the structure so it drains and aerates better. So it helps both beds. It does improve sandy soil's retention, so 'only the clay' is wrong. Compost opens pores and feeds soil rather than sealing it. And it cannot transmute sand into clay or clay into sand — it moderates each toward a better loam-like balance.

34. (B) Only a direct count of micro-organisms, mites and worms shows living things. Bubbling in water shows trapped air, not life. Misting on a heated tube shows moisture (condensed water vapour), not breathing. Mass loss on strong heating is mostly evaporated water plus burnt humus — humus is dead organic matter, so the loss does not by itself prove living organisms. Each non-life option misreads a physical property as evidence of life.

35. (D) Soil organisms need moisture to live, humus as food, and air for respiration — all abundant in loam, scarce in salty sand. Organisms do not favour the largest particles; loam's value is its balanced structure with humus, not coarse grains. Fast drainage is not what supports life — water and food do. Cold does not maximise life; many organisms need moderate warmth.

36. (B) Clay's fine particles pack so tightly that its pores are tiny and resist water flow — that is why a clay bowl holds water and why clay soil drains slowly. Tight packing slows flow, it does not force water straight through fast. Clay actually holds a lot of water (high water-holding capacity), not none. And clay's percolation rate is far lower than sand's despite both being mineral — particle size and packing, not just composition, set the rate.

37. (A) Standing water saturates clay's tiny pores, shutting out air, so gram roots are starved of oxygen and rot — 'more water' harms an aeration-loving crop. Clay drains slowly, so plot B did not dry out from fast drainage. Flooding leaches nutrients downward, not upward to the surface. Gram is not a water crop; plot A, which thrived, was the correctly managed one, so the standing-water plot is the failure, not the reverse.

38. (D) Paddy needs standing water, so it suits the water-logging clay; gram favours moisture-holding clayey-loam; cotton prefers a light, well-drained sandy-loam; groundnut wants loose, fast-draining sand for its pods. The other matchings put paddy in fast-draining soil (it would dry out), groundnut in water-logging clay (its pods rot), or cram all crops into clay despite their differing water needs.

39. (D) Soil K's intermediate drainage holds steady moisture yet sheds the surplus, matching tomatoes' need to stay moist but not water-logged. Soil J drains so fast it dries out by noon — too dry. Soil L stays soggy for days, water-logging the roots; 'wettest is best' ignores that standing water starves roots of air. Slow drainage does not guarantee air — it causes the opposite by keeping pores water-filled.

40. (C) Field 1's coarse sand drains so fast that watered moisture sinks past the roots, causing drought wilting; field 2's heavy clay holds water and excludes air, water-logging and rotting the roots — two opposite causes. Humus is not the culprit in either, and field 1 is not water-logged. Sand drains (it does not water-log) and clay retains water (it does not dry out), so swapping their behaviours is wrong. High water-holding capacity keeps soil moist, the opposite of drying crops out.

41. (D) No single soil is best for all crops — match the soil to the crop's water needs. Fast-draining sand suits groundnut and other drought-tolerant crops; water-holding clay suits paddy and other water-lovers; loam suits most general crops. So both sweeping claims fail. The third option is also self-contradictory, since fastest drainage and highest retention cannot coexist in one soil.

42. (C) Cotton favours lighter, well-drained sandy-loam, so it belongs in field G; paddy (needs standing water) and gram suit the moisture-holding clayey-loam of field H. Paddy in the fast-draining field G would lose its needed water. Soil type matters, so 'all crops in G equally' is false. Cotton prefers light, drained soil, not water-logged clay, so placing it in field H is wrong.

43. (B) Soil pollution is the addition of harmful substances; toxic effluent (i) and plastic waste (ii) both fit, poisoning or choking the soil. Runoff stripping topsoil (iii) is erosion — removal, not contamination. Leaves rotting into humus (iv) is natural enrichment that improves soil. So only (i) and (ii) are pollution. Calling erosion 'pollution' confuses removal with contamination; calling beneficial humus 'pollution' is wrong; and not every change is pollution — only harmful additions are.

44. (A) Up-and-down furrows act as channels that speed runoff and cut gullies (farmer 1). Level terraces and contour ploughing interrupt and slow the downhill flow so water soaks in, holding soil in place (farmer 2). Farmer 2's method slows water, not speeds it. It does not rely on evaporation. And terracing protects the soil in place rather than removing it beforehand.

45. (C) Wind erosion on a flat plain is best slowed by shelter belts (tree rows that break the wind); water erosion on a steep slope is best fought by terraces and contour ploughing that slow runoff. Swapping them is wrong — terraces suit slopes, not flat ground, and shelter belts suit wind, not gullies. Deep ploughing loosens soil and can worsen erosion, not stop it. Leaving sites bare maximises both wind and water erosion.

46. (D) Gap (pore) size set by particle size governs flow: large marbles leave wide gaps (fast, like sand); fine grains leave narrow gaps (slow, like clay). Marbles do not dissolve to open channels. Fine grains do not push water upward against gravity. And the marbles clearly have gaps between them — wide ones — which is exactly why water runs straight through, so 'no gaps' is the opposite of the truth.

47. (B) Weathering is the in-place break-up of rock — here the frost cracking the boulder over decades. Erosion is the carrying-away of loosened material — here the flood sweeping the rubble to the valley. So only the flood stage is erosion; labelling the whole event erosion ignores the in-place break-up. Calling it all weathering ignores the transport. Reversing the two mislabels both. And weathering and erosion are not restricted to wind — frost and water do both jobs.

48. (B) Identical granite means climate decides: the hot humid valley weathers faster and its lush vegetation drops litter that becomes deep humus, so its soil is thicker. Cold dry air slows, not speeds, weathering, so the ridge prediction is backward. A high windy position favours erosion and bare rock, not deep soil. Same rock does not force same soil — environment makes the depths differ markedly.

49. (A) Freeze-thaw is a powerful physical weathering agent: water seeps into cracks, freezes, expands and pries the rock apart, repeated each winter — so the riverbed chunk (1) crumbles most. The sealed dry case lacks the water needed for freeze-thaw, so little happens. Dry buried desert sand provides no freezing water, and burial alone does not crush sandstone to bits. Steady dry indoor air offers neither moisture nor freezing, so weathering is minimal. Water plus freezing drives the break-up, present only in the riverbed.

50. (D) Soil = weathered mineral grains + humus + living organisms; bedrock = solid unweathered rock with essentially none of that life. The first distractor reverses which layer holds life. Bedrock is not 'equally full' of humus and organisms — soil holds them, bedrock does not. And soil is not pure organic matter; it is mostly weathered mineral material, so 'no mineral content' is false.

51. (B) The paired change — A horizon paling and thinning, B horizon gaining clay and iron — is the signature of leaching: rainwater seeps down, strips fine clay and dissolved iron from the A horizon (zone of loss), and deposits them in the B horizon (zone of gain). Capillary rise from bedrock moves upward and would not pale the A horizon. Wind does

not selectively load a buried horizon. Soil cannot manufacture clay and iron from trapped air; the material was transferred from above.

52. (C) Rates: $S = 360 / 4 = 90$; $U = 360 / 6 = 60$; $T = 360 / 12 = 30$ mL/min. Sandiest = fastest = S, then U, then T (clayiest). U is not sandiest — S drains faster. The longest time (T) is the clayiest, not the loosest. Equal volume with different times means different rates, so the textures differ.

53. (B) Rate = $240 / 6 = 40$ mL/min. Time for 600 mL = $600 / 40 = 15$ min. The 24000 figure wrongly multiplies 40×600 instead of dividing. The 1440 mL/min figure multiplies 240×6 instead of dividing. The 6 mL/min figure mistakes the time (6 min) for the rate. Only rate = volume/time and time = volume/rate give the right pair.

54. (D) The soil is the SAME in both; halving the depth halves the path and roughly halves the time (5 vs 10 min), so the shorter time reflects depth, not texture. Longer time does not prove coarser grains. Depth clearly affects drain time here, so saying it has 'no effect' is false. A faster time does not always mean larger particles when depth differs — that is exactly the confound being exposed.

55. (D) A-horizon thickness tracks organic input minus loss. Forest (plot 1) adds the most litter and is never disturbed, so it is thickest; mulched, rotated cropland (plot 2) keeps some cover and organic matter, so it is intermediate; bare cropping with burned stubble (plot 3) strips organic matter and exposes soil to erosion, so it is thinnest. Cropping and burning destroy, not build, topsoil. Same climate and rock do not force equal A horizons — management differs. Mulched cropland helps but does not out-build undisturbed forest.

56. (D) Mulch absorbs the impact of falling rain and shelters the surface from wind, and it conserves moisture beneath — so plot 2 kept its topsoil and stayed damper. Mulch does not make soil heavier. It does not seal off percolation — water still soaks through. And it cannot turn topsoil into bedrock; it simply shields and conserves the surface.

57. (C) Trampling compacts the soil, collapsing its pore spaces, so water cannot seep down — hence puddles and a LOWER percolation rate on the path than the loose lawn. Compaction slows, not speeds, drainage, so a higher rate is wrong. The same soil can differ in drainage once compacted, so 'identical' is wrong. Compaction shrinks pores, the opposite of enlarging them, so that reasoning contradicts the puddling.

58. (B) Weathering that makes soil parent material is slow: lichens and roots prising grains (i), freeze-thaw wedging cracks (ii), and wind-blown sand abrading the face (iv) all fragment rock over long times. A one-second lightning char (iii) is a momentary surface event, not a soil-building weathering agent. Weathering is not confined to sudden violent events — slow steady processes dominate. And rock does not turn to soil by melting; it weathers gradually.

59. (A) Top to bottom: O (leaf litter), A (dark crumbly humus topsoil), B (pale clay-stained accumulation band), C (loose rock chips = parent rock), bedrock (solid rock). So the pale stained band is correctly the B horizon. The dark crumbly band is the A horizon, not C. The loose rock chips are the C horizon, not the humus-rich A. The half-rotted leaves are the O horizon, not bedrock.

60. (C) Rate = volume / time. Soil X: $720 / 9 = 80$ mL/min. Soil Y: $720 / 15 = 48$ mL/min. So X is the faster drainer, and Y's longer time means it is slower, not faster — a longer clock for the same volume signals a lower rate. Equal volume with different times gives unequal rates, so they are not equal. Multiplying $720 \times 9 (= 6480)$ misuses the formula; rate is volume divided by time, not multiplied.